

Tangents to a circle



- 1
 - a $OB < OC$
 - b Point B
 - c The angle is a right angle.
 - d $\angle OBA$ is a right angle
- 2
 - a No solution provided.
 - b They are collinear.
- 3
 - a $x = 90^\circ$ (A tangent to a circle is perpendicular to the radius drawn to the point of contact)
 - b $x = 63^\circ$ (The interior angle sum of a triangle is 180°)
 - c $x = 40$ (Two straight lines are tangents issued from the same external point)
 - d $x = 59^\circ, y = 59^\circ$ (The angle between a tangent and a chord drawn to the point of contact is equal to the angle in the alternate segment)
 - e $x = 24$ (The interior angle sum of a triangle is 180°)
 - f $x = 131^\circ$ (The interior angle sum of a quadrilateral is 360°)
- 4 136 units
- 5 $\angle SUV = 48^\circ$
- 6 $OK = 21$
- 7
 - a $y = 53^\circ$
 - b $x = 37^\circ$
- 8
 - a $x = 55^\circ$
 - b $x = 63^\circ$
 - c $a = 34^\circ$
 - d $k = 63^\circ$
 - e $n = 42^\circ$
 - f $y = 72.5^\circ$
- 9
 - a $x = 8$
 - b $y = 65^\circ$
- 10 $\angle YZT = 57^\circ$
- 11 $\angle APB = 44^\circ$
- 12 $b = 72^\circ$

13 $p = 34 \text{ cm}$



14 $f = 68^\circ$

Proofs

15 No solution provided.

16 No solution provided.

17 No solution provided.